Unsupervised Learning — Problem Set

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1. Cluster Analysis

Problem 1.1 (Distance matrix, grouping, centroid and SSE).
Consider the five 2-D points

$$P_1 = (1, 1), \quad P_2 = (2, 1), \quad P_3 = (1, 2), \quad P_4 = (8, 8), \quad P_5 = (9, 7).$$

(a) Compute the pairwise Euclidean distance matrix (show distances to two decimal places).

(b) Based on the distances, suggest a natural grouping (clusters) and list which points belong to each cluster.

(c) For each proposed cluster, compute the centroid (mean point).

(d) Compute the Sum of Squared Errors (SSE) for the clustering you proposed.

Problem 1.2 (Manual k-means clustering to convergence).
Consider the four points

$$Q_1 = (1, 1), \quad Q_2 = (1, 2), \quad Q_3 = (4, 4), \quad Q_4 = (5, 4).$$

Run the k-means algorithm with $k = 2$ and the following initial centroids:

$$\mu_1^{(0)} = (1, 1), \qquad \mu_2^{(0)} = (5, 4).$$

Perform full iterations (assignment step and centroid update) until the centroids do not change. Show all steps and final clusters.

2. Definition of Clusters

Problem 2.1 Given points $P(0, 0)$, $Q(0, 1)$, $R(5, 5)$ and $S(6, 5)$, determine one valid interpretation of clusters based on density.

Problem 2.2 State one difference between a "compactness-based" cluster and a "connectivity-based" cluster.

3. Proximity Measures for Discrete Variables

Problem 3.1 (Hamming distance and simple matching coefficient).
Two customers rated four products as `Like (1)` or `Dislike (0)`:

$$A = (1, 0, 1, 1), \qquad B = (1, 1, 0, 1).$$

(a) Compute the Hamming distance between A and B.

(b) Compute the Simple Matching Coefficient (SMC).

Problem 3.2 (Jaccard similarity and Jaccard distance).
Two items have binary attribute vectors representing the presence (1) or absence (0) of features:

$$X = (1, 1, 0, 1, 0), \qquad Y = (1, 0, 0, 1, 1).$$

(a) Compute the Jaccard similarity.

(b) Compute the Jaccard distance.

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4. Proximity Measures for Mixed Variables

Problem 4.1 (Gower similarity for one numeric and one categorical variable).
Consider two objects with attributes Age (numeric, range 0–100) and Color (categorical):

$$O_1 = (20,\ \text{Red}), \qquad O_2 = (35,\ \text{Red}).$$

Compute the Gower similarity.

Problem 4.2 (Gower distance with mixed numeric, binary, and categorical variables).
Three attributes: 1. Height (numeric, range 100–200 cm) 2. Smoker (binary 0/1) 3. Eye Color (categorical: Brown/Blue/Green)

$$A = (160,\ 1,\ \text{Brown}), \qquad B = (190,\ 0,\ \text{Blue}).$$

Compute the Gower similarity and Gower distance.

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5. Clustering Criteria

Problem 5.1 (Compute Within-Cluster and Between-Cluster Scatter).
Three 2D points are assigned to two clusters:

$$C_1 = \{(1, 1), (2, 1)\}, \qquad C_2 = \{(4, 3)\}.$$

Compute:

(a) Within-cluster scatter W using squared Euclidean distance.

(b) Between-cluster scatter B using the distance between centroids and the overall mean.

Problem 5.2 (Compute Dunn Index).
Two clusters:

$$C_1 = \{(0,0),(2,0)\}, \qquad C_2 = \{(5,0),(7,0)\}.$$

Compute the Dunn Index using:

$$\mathrm{DI} = \frac{\min_{i \neq j} d(C_i, C_j)}{\max_k \mathrm{diam}(C_k)}.$$

6. Hierarchical Clustering

Problem 6.1 (Single-Linkage Hierarchical Clustering).
Four points on a line:

$$A = 1, \quad B = 2, \quad C = 5, \quad D = 9.$$

Perform hierarchical clustering using **single linkage**. Show all merging steps and distances.

Problem 6.2 (Complete-Linkage Clustering).
Use the same points:

$$A = 1, \quad B = 2, \quad C = 5, \quad D = 9.$$

Perform hierarchical clustering using **complete linkage**.

7. Agglomerative and Divisive Clustering

A. Agglomerative Clustering :

Problem 7A.1 (Agglomerative clustering with single linkage).
Given the points

$$A = (1,1),\ B = (2,1),\ C = (6,4),\ D = (7,5),$$

perform agglomerative hierarchical clustering using **single linkage**. Show all merging steps.

Problem 7A.2 (Agglomerative clustering with complete linkage).
Use the same points

$$A = (1,1),\ B = (2,1),\ C = (6,4),\ D = (7,5)$$

but perform hierarchical clustering using **complete linkage**.

B. Divisive Clustering :

Problem 7B.1 (Top-down splitting by distance).
Points:

$$A = (0, 0),\ B = (1, 0),\ C = (5, 4),\ D = (6, 5).$$

Perform a basic **divisive** clustering: 1. Start with one cluster. 2. Split cluster using the pair of points with largest distance.

Problem 7B.2 (Divisive clustering via SSE splitting).
Points:

$$P_1 = (1, 2),\ P_2 = (2, 1),\ P_3 = (8, 9),\ P_4 = (9, 8).$$

Use divisive clustering: 1. Start with all points in one cluster. 2. Split using 2-means into two groups.

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8. Fuzzy Clustering

Problem 8.1 (Compute membership values in Fuzzy C-Means).
Two cluster centers:

$$C_1 = (1, 1), \quad C_2 = (5, 5),$$

and a point

$$X = (2, 2).$$

Use fuzziness parameter $m = 2$. Compute the membership values u_1 and u_2.

Problem 8.2 (Update fuzzy cluster center).
Cluster 1 center update formula:

$$C_1 = \frac{\sum u_{1j}^m x_j}{\sum u_{1j}^m}.$$

Given:

$$x_1 = (0, 0),\ x_2 = (2, 2),$$

$$u_{11} = 0.8,\ u_{12} = 0.4, \quad m = 2.$$

Compute C_1.

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9. Cluster Validity

Problem 9.1 (Silhouette score).
Three points in a cluster:

$$A = (0, 0), B = (1, 0), C = (10, 0).$$

Clusters:

$$C_1 = \{A, B\}, \quad C_2 = \{C\}.$$

Compute silhouette value for point A.

Problem 9.2 (Davies–Bouldin Index).

Cluster centers:

$$C_1 = (1, 1), \quad C_2 = (5, 5),$$

Cluster scatters:

$$S_1 = 1, \quad S_2 = 2.$$

$$DBI = \frac{1}{k} \sum_i \max_{j \neq i} \frac{S_i + S_j}{d(C_i, C_j)}.$$

Compute DBI.

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10. Clustering Applications

Problem 10.1 (Image compression using clustering).
A small 4-pixel image has RGB color points:

$$(255, 0, 0),\ (250, 10, 5),\ (0, 255, 0),\ (5, 240, 10).$$

Explain how k-means clustering with $k = 2$ can be used for image compression and identify the two natural color clusters.

Problem 10.2 (Anomaly detection in banking).
A bank records daily transaction amounts (in thousands):

$$1.2,\ 1.0,\ 1.3,\ 0.9,\ 12.0.$$

Explain how clustering helps reveal anomalies and identify which point is abnormal.

Problem 10.3 (Recommender systems).
A movie platform represents users with feature vectors:

$$U_1 = (4, 5),\ U_2 = (5, 4),\ U_3 = (1, 1),\ U_4 = (2, 1).$$

Coordinates represent *(average rating of action movies, average rating of comedies)*. Describe the clusters and explain how they help produce recommendations.

Problem 10.4 (Medical data categorization).
A hospital records patient features:

$$\text{Blood Pressure (BP)}, \quad \text{Cholesterol (C)}.$$

Given data:

$$(120, 180),\ (118, 175),\ (160, 240),\ (165, 250).$$

Identify natural clusters and explain how clustering supports diagnosis.

Problem 10.5 (Geographical hotspot detection).
A city tracks accident locations on a 2D map:

$$(2, 3),\ (3, 2),\ (20, 25),\ (22, 24),\ (19, 26).$$

Identify accident hotspots.

Solutions

Solution 1.1

(a) Pairwise Euclidean distances. Recall the Euclidean distance between (x_i, y_i) and (x_j, y_j) is

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}.$$

We compute only the upper triangle (matrix is symmetric).

$$\begin{aligned}
d(P_1, P_2) &= \sqrt{(1-2)^2 + (1-1)^2} = \sqrt{1+0} = 1.00,\\
d(P_1, P_3) &= \sqrt{(1-1)^2 + (1-2)^2} = \sqrt{0+1} = 1.00,\\
d(P_1, P_4) &= \sqrt{(1-8)^2 + (1-8)^2} = \sqrt{49+49} = \sqrt{98} \approx 9.90,\\
d(P_1, P_5) &= \sqrt{(1-9)^2 + (1-7)^2} = \sqrt{64+36} = \sqrt{100} = 10.00,\\
d(P_2, P_3) &= \sqrt{(2-1)^2 + (1-2)^2} = \sqrt{1+1} = \sqrt{2} \approx 1.41,\\
d(P_2, P_4) &= \sqrt{(2-8)^2 + (1-8)^2} = \sqrt{36+49} = \sqrt{85} \approx 9.22,\\
d(P_2, P_5) &= \sqrt{(2-9)^2 + (1-7)^2} = \sqrt{49+36} = \sqrt{85} \approx 9.22,\\
d(P_3, P_4) &= \sqrt{(1-8)^2 + (2-8)^2} = \sqrt{49+36} = \sqrt{85} \approx 9.22,\\
d(P_3, P_5) &= \sqrt{(1-9)^2 + (2-7)^2} = \sqrt{64+25} = \sqrt{89} \approx 9.43,\\
d(P_4, P_5) &= \sqrt{(8-9)^2 + (8-7)^2} = \sqrt{1+1} = \sqrt{2} \approx 1.41.
\end{aligned}$$

So the symmetric distance matrix (rounded) is:

	P_1	P_2	P_3	P_4	P_5
P_1	0	1.00	1.00	9.90	10.00
P_2	1.00	0	1.41	9.22	9.22
P_3	1.00	1.41	0	9.22	9.43
P_4	9.90	9.22	9.22	0	1.41
P_5	10.00	9.22	9.43	1.41	0

(b) Natural grouping. Observe small distances:

$$d(P_1, P_2) = 1.00, \quad d(P_1, P_3) = 1.00, \quad d(P_2, P_3) \approx 1.41,$$

and similarly

$$d(P_4, P_5) \approx 1.41,$$

while distances between any of $\{P_1, P_2, P_3\}$ and any of $\{P_4, P_5\}$ are about 9.2–10. Thus the natural grouping (two clusters) is:

$$\text{Cluster A} = \{P_1, P_2, P_3\}, \qquad \text{Cluster B} = \{P_4, P_5\}.$$

(c) Centroids of clusters. Centroid is the mean of coordinates.
Cluster A (P_1, P_2, P_3):

$$\bar{x}_A = \frac{1+2+1}{3} = \frac{4}{3} \approx 1.333, \qquad \bar{y}_A = \frac{1+1+2}{3} = \frac{4}{3} \approx 1.333.$$

So centroid $C_A = (4/3,\, 4/3) \approx (1.33, 1.33)$.

Cluster B (P_4, P_5):

$$\bar{x}_B = \frac{8+9}{2} = 8.5, \qquad \bar{y}_B = \frac{8+7}{2} = 7.5.$$

So centroid $C_B = (8.5,\ 7.5)$.

(d) Sum of Squared Errors (SSE). SSE is the sum of squared Euclidean distances of points to their cluster centroid.

For Cluster A:

$$\begin{aligned}
d^2(P_1, C_A) &= (1 - \tfrac{4}{3})^2 + (1 - \tfrac{4}{3})^2 = \left(-\tfrac{1}{3}\right)^2 + \left(-\tfrac{1}{3}\right)^2 = \tfrac{1}{9} + \tfrac{1}{9} = \tfrac{2}{9} \approx 0.222,\\
d^2(P_2, C_A) &= (2 - \tfrac{4}{3})^2 + (1 - \tfrac{4}{3})^2 = \left(\tfrac{2}{3}\right)^2 + \left(-\tfrac{1}{3}\right)^2 = \tfrac{4}{9} + \tfrac{1}{9} = \tfrac{5}{9} \approx 0.556,\\
d^2(P_3, C_A) &= (1 - \tfrac{4}{3})^2 + (2 - \tfrac{4}{3})^2 = \left(-\tfrac{1}{3}\right)^2 + \left(\tfrac{2}{3}\right)^2 = \tfrac{1}{9} + \tfrac{4}{9} = \tfrac{5}{9} \approx 0.556.
\end{aligned}$$

Sum for Cluster A:

$$SSE_A = \tfrac{2}{9} + \tfrac{5}{9} + \tfrac{5}{9} = \tfrac{12}{9} = \tfrac{4}{3} \approx 1.333.$$

For Cluster B:

$$\begin{aligned}
d^2(P_4, C_B) &= (8 - 8.5)^2 + (8 - 7.5)^2 = (-0.5)^2 + (0.5)^2 = 0.25 + 0.25 = 0.50,\\
d^2(P_5, C_B) &= (9 - 8.5)^2 + (7 - 7.5)^2 = (0.5)^2 + (-0.5)^2 = 0.25 + 0.25 = 0.50.
\end{aligned}$$

Sum for Cluster B:

$$SSE_B = 0.50 + 0.50 = 1.00.$$

Total SSE:

$$SSE_{\text{total}} = SSE_A + SSE_B = \tfrac{4}{3} + 1 = \tfrac{7}{3} \approx 2.333.$$

Conclusion: The two clusters are compact (small intra-cluster distances) and well separated (large inter-cluster distances), which supports the chosen grouping.

Solution 1.2

Initialization (given):

$$\mu_1^{(0)} = (1, 1), \qquad \mu_2^{(0)} = (5, 4).$$

Iteration 1 — Assignment step. We assign each point to the nearest centroid using Euclidean distance.

Distances to $\mu_1^{(0)} = (1, 1)$:

$$\begin{aligned}
d(Q_1, \mu_1^{(0)}) &= \sqrt{(1-1)^2 + (1-1)^2} = 0,\\
d(Q_2, \mu_1^{(0)}) &= \sqrt{(1-1)^2 + (2-1)^2} = 1,\\
d(Q_3, \mu_1^{(0)}) &= \sqrt{(4-1)^2 + (4-1)^2} = \sqrt{9+9} = \sqrt{18} \approx 4.24,\\
d(Q_4, \mu_1^{(0)}) &= \sqrt{(5-1)^2 + (4-1)^2} = \sqrt{16+9} = \sqrt{25} = 5.
\end{aligned}$$

Distances to $\mu_2^{(0)} = (5, 4)$:

$$\begin{aligned}
d(Q_1, \mu_2^{(0)}) &= \sqrt{(1-5)^2 + (1-4)^2} = \sqrt{16+9} = \sqrt{25} = 5,\\
d(Q_2, \mu_2^{(0)}) &= \sqrt{(1-5)^2 + (2-4)^2} = \sqrt{16+4} = \sqrt{20} \approx 4.47,\\
d(Q_3, \mu_2^{(0)}) &= \sqrt{(4-5)^2 + (4-4)^2} = \sqrt{1+0} = 1,\\
d(Q_4, \mu_2^{(0)}) &= \sqrt{(5-5)^2 + (4-4)^2} = 0.
\end{aligned}$$

Compare distances and assign to the nearest centroid:

$$\begin{aligned}
Q_1 &: d(\cdot,\mu_1) = 0 < d(\cdot,\mu_2) = 5 \quad \Rightarrow \; Q_1 \in C_1, \\
Q_2 &: d(\cdot,\mu_1) = 1 < d(\cdot,\mu_2) \approx 4.47 \quad \Rightarrow \; Q_2 \in C_1, \\
Q_3 &: d(\cdot,\mu_1) \approx 4.24 > d(\cdot,\mu_2) = 1 \quad \Rightarrow \; Q_3 \in C_2, \\
Q_4 &: d(\cdot,\mu_1) = 5 > d(\cdot,\mu_2) = 0 \quad \Rightarrow \; Q_4 \in C_2.
\end{aligned}$$

So after assignment:

$$C_1^{(1)} = \{Q_1, Q_2\}, \qquad C_2^{(1)} = \{Q_3, Q_4\}.$$

Iteration 1 — Update centroids. Compute new centroids as the mean of assigned points. For $C_1^{(1)}$:

$$\mu_1^{(1)} = \Big(\frac{1+1}{2}, \frac{1+2}{2}\Big) = (1, 1.5).$$

For $C_2^{(1)}$:

$$\mu_2^{(1)} = \Big(\frac{4+5}{2}, \frac{4+4}{2}\Big) = (4.5, 4).$$

Check for convergence: $\mu_1^{(1)} \neq \mu_1^{(0)}$ and $\mu_2^{(1)} \neq \mu_2^{(0)}$. Continue.

Iteration 2 — Assignment step with updated centroids.
Distances to $\mu_1^{(1)} = (1, 1.5)$:

$$\begin{aligned}
d(Q_1, \mu_1^{(1)}) &= \sqrt{(1-1)^2 + (1-1.5)^2} = 0.5, \\
d(Q_2, \mu_1^{(1)}) &= \sqrt{(1-1)^2 + (2-1.5)^2} = 0.5, \\
d(Q_3, \mu_1^{(1)}) &= \sqrt{(4-1)^2 + (4-1.5)^2} = \sqrt{9 + 6.25} = \sqrt{15.25} \approx 3.90, \\
d(Q_4, \mu_1^{(1)}) &= \sqrt{(5-1)^2 + (4-1.5)^2} = \sqrt{16 + 6.25} = \sqrt{22.25} \approx 4.72.
\end{aligned}$$

Distances to $\mu_2^{(1)} = (4.5, 4)$:

$$\begin{aligned}
d(Q_1, \mu_2^{(1)}) &= \sqrt{(1-4.5)^2 + (1-4)^2} = \sqrt{12.25 + 9} = \sqrt{21.25} \approx 4.61, \\
d(Q_2, \mu_2^{(1)}) &= \sqrt{(1-4.5)^2 + (2-4)^2} = \sqrt{12.25 + 4} = \sqrt{16.25} \approx 4.03, \\
d(Q_3, \mu_2^{(1)}) &= \sqrt{(4-4.5)^2 + (4-4)^2} = \sqrt{0.25 + 0} = 0.5, \\
d(Q_4, \mu_2^{(1)}) &= \sqrt{(5-4.5)^2 + (4-4)^2} = \sqrt{0.25 + 0} = 0.5.
\end{aligned}$$

Now assign to nearest centroid:

$$\begin{aligned}
Q_1 &: d(\cdot,\mu_1) \approx 0.5 < d(\cdot,\mu_2) \approx 4.61 \Rightarrow Q_1 \in C_1, \\
Q_2 &: d(\cdot,\mu_1) \approx 0.5 < d(\cdot,\mu_2) \approx 4.03 \Rightarrow Q_2 \in C_1, \\
Q_3 &: d(\cdot,\mu_1) \approx 3.90 > d(\cdot,\mu_2) = 0.5 \Rightarrow Q_3 \in C_2, \\
Q_4 &: d(\cdot,\mu_1) \approx 4.72 > d(\cdot,\mu_2) = 0.5 \Rightarrow Q_4 \in C_2.
\end{aligned}$$

Assignments did not change from the previous iteration:

$$C_1^{(2)} = \{Q_1, Q_2\}, \qquad C_2^{(2)} = \{Q_3, Q_4\}.$$

Iteration 2 — Update centroids. Recompute centroids (they will be the same as in iteration 1):

$$\mu_1^{(2)} = (1, 1.5), \qquad \mu_2^{(2)} = (4.5, 4).$$

Convergence: The centroids did not change from iteration 1 to iteration 2, so the algorithm has converged.

Final clusters and centroids:

$$C_1^* = \{Q_1, Q_2\}, \quad \mu_1^* = (1, 1.5); \qquad C_2^* = \{Q_3, Q_4\}, \quad \mu_2^* = (4.5, 4).$$

Solution 2.1

Density-based definition groups points that have nearby neighbors. Distances:

$$d(P, Q) = 1, \quad d(R, S) = 1.$$

Distances across groups:

$$d(P, R) = \sqrt{50} \approx 7.07.$$

Thus two dense groups exist: $\{P, Q\}$ and $\{R, S\}$.

Solution 2.2

A compactness-based cluster groups points that are close to a centroid (e.g., k-means). A connectivity-based cluster groups points linked by minimum distances often forming hierarchical structures. Thus compactness uses global centroid structure; connectivity uses local neighbor chains.

Solution 3.1

(a) Hamming distance. Count positions where A and B differ:

$$\begin{aligned} 1 = 1 \quad (0), \\ 0 \neq 1 \quad (1), \\ 1 \neq 0 \quad (1), \\ 1 = 1 \quad (0). \end{aligned}$$

Total differences = 2.

$$d_H(A, B) = 2.$$

(b) Simple Matching Coefficient. Number of matches = 2 (positions 1 and 4). Total attributes = 4.

$$SMC = \frac{\text{matches}}{\text{total}} = \frac{2}{4} = 0.5.$$

Solution 3.2

Intersection = number of positions where both are 1:

$$X \cap Y = \{1, 4\} \Rightarrow 2.$$

Union = number of positions where at least one is 1:

$$X \cup Y = \{1, 2, 4, 5\} \Rightarrow 4.$$

(a) Jaccard similarity:

$$J(X, Y) = \frac{2}{4} = 0.5.$$

(b) Jaccard distance:

$$d_J(X, Y) = 1 - J(X, Y) = 1 - 0.5 = 0.5.$$

Solution 4.1

Numeric similarity:

$$s_{\text{num}} = 1 - \frac{|20 - 35|}{100} = 1 - \frac{15}{100} = 0.85.$$

Categorical similarity (1 if equal, 0 otherwise):

$$s_{\text{cat}} = 1.$$

Gower similarity:

$$S = \frac{s_{\text{num}} + s_{\text{cat}}}{2} = \frac{0.85 + 1}{2} = 0.925.$$

Solution 4.2

1. Numeric attribute — Height:

$$s_1 = 1 - \frac{|160 - 190|}{200 - 100} = 1 - \frac{30}{100} = 0.70.$$

2. Binary attribute — Smoker:

$$s_2 = 0 \quad \text{(values differ)}.$$

3. Categorical attribute — Eye Color:

$$s_3 = 0 \quad \text{(Brown} \neq \text{Blue)}.$$

Gower similarity:

$$S = \frac{s_1 + s_2 + s_3}{3} = \frac{0.70 + 0 + 0}{3} = \frac{0.70}{3} \approx 0.233.$$

Gower distance:

$$D = 1 - S = 1 - 0.233 = 0.767.$$

Solution 5.1

Cluster centroids:

$$\mu_1 = (1.5, 1), \qquad \mu_2 = (4, 3).$$

Overall mean:

$$\mu = \left(\frac{1+2+4}{3}, \frac{1+1+3}{3}\right) = (2.33,\ 1.67).$$

(a) Within-cluster scatter.
Cluster 1:

$$(1-1.5)^2 + (1-1)^2 = 0.25$$
$$(2-1.5)^2 + (1-1)^2 = 0.25$$
$$W_1 = 0.25 + 0.25 = 0.50$$

Cluster 2:

$$W_2 = 0 \quad \text{(only one point)}$$

Total:

$$W = W_1 + W_2 = 0.50.$$

(b) Between-cluster scatter.
Using:

$$B = \sum_i n_i \|\mu_i - \mu\|^2$$

Cluster 1 ($n_1 = 2$):

$$\|\mu_1 - \mu\|^2 = (1.5 - 2.33)^2 + (1 - 1.67)^2 = 0.6889 + 0.4489 = 1.1378.$$
$$B_1 = 2 \times 1.1378 = 2.2756$$

Cluster 2 ($n_2 = 1$):

$$\|\mu_2 - \mu\|^2 = (4 - 2.33)^2 + (3 - 1.67)^2 = 2.7889 + 1.7556 = 4.5445.$$
$$B_2 = 4.5445$$

Total:

$$B = 2.2756 + 4.5445 = 6.8201.$$

Solution 5.2

Cluster diameters (max distance within a cluster).
Cluster 1:

$$\mathrm{diam}(C_1) = |0 - 2| = 2.$$

Cluster 2:

$$\mathrm{diam}(C_2) = |5 - 7| = 2.$$

$$\max_k \mathrm{diam}(C_k) = 2.$$

Inter-cluster distance = min distance between points in different clusters.
Pairs:

$$|2-5| = 3, \quad |2-7| = 5, \quad |0-5| = 5, \quad |0-7| = 7.$$

Minimum = 3.

$$\mathrm{DI} = \frac{3}{2} = 1.5.$$

Solution 6.1

Pairwise distances:

$$d(A,B)=1,\ d(B,C)=3,\ d(C,D)=4,\ d(A,C)=4,\ d(A,D)=8,\ d(B,D)=7.$$

Step 1. Merge closest pair:

$$A,B \quad \text{(distance 1)}$$

Clusters:

$$\{AB\},\ \{C\},\ \{D\}$$

Step 2. Compute distances using single-linkage:

$$d(AB,C)=\min(d(A,C),d(B,C))=\min(4,3)=3$$

$$d(AB,D)=\min(d(A,D),d(B,D))=\min(8,7)=7$$

$$d(C,D)=4$$

Merge $\{AB\}$ and C (distance 3).
Clusters:

$$\{ABC\},\ \{D\}$$

Step 3.

$$d(ABC,D)=\min(d(A,D),d(B,D),d(C,D))=\min(8,7,4)=4.$$

Merge.
Final cluster:

$$\{ABCD\}$$

Merging distances:

$$1,\ 3,\ 4.$$

Solution 6.2

Pairwise distances (same as before):

$$d(A,B)=1,\ d(B,C)=3,\ d(C,D)=4,\ d(A,C)=4,\ d(A,D)=8,\ d(B,D)=7.$$

Step 1. Merge closest pair:

$$A,B \quad \text{(distance 1)}$$

Clusters:

$$\{AB\},\ \{C\},\ \{D\}$$

Step 2. Compute complete-link distances:

$$d(AB,C)=\max(d(A,C),d(B,C))=\max(4,3)=4$$

$$d(AB,D)=\max(d(A,D),d(B,D))=\max(8,7)=8$$

$$d(C,D)=4$$

Next merge (tie at 4): Merge $\{C\}$ and $\{D\}$.
Clusters:

$$\{AB\},\ \{CD\}$$

Step 3. Compute distance between AB and CD:

$$d(AB,CD)=\max(d(A,C),d(A,D),d(B,C),d(B,D))$$

$$= \max(4, 8, 3, 7) = 8.$$

Merge.
Final cluster:

$$\{ABCD\}$$

Merging distances:

$$1,\ 4,\ 8.$$

Solution 7A.1

Compute pairwise distances:

$$d(A, B) = 1,\ d(A, C) = 5,\ d(A, D) = 7,$$

$$d(B, C) = \sqrt{(2-6)^2 + (1-4)^2} = 5, \quad d(B, D) = \sqrt{(2-7)^2 + (1-5)^2} = 6,$$
$$d(C, D) = \sqrt{(6-7)^2 + (4-5)^2} = \sqrt{2} = 1.41.$$

Step 1: Merge closest pair:

$$A, B \quad (d = 1)$$

Clusters: {AB}, {C}, {D}.
Distances using single linkage:

$$d(AB, C) = \min(d(A, C), d(B, C)) = \min(5, 5) = 5,$$

$$d(AB, D) = \min(7, 6) = 6,$$

$$d(C, D) = 1.41.$$

Step 2: Merge C and D (distance 1.41). Clusters: {AB}, {CD}.
Distance between AB and CD:

$$d(AB, CD) = \min(5, 6, 5, 7) = 5.$$

Step 3: Merge {AB} and {CD}. Final cluster: {ABCD}.
Merging distances: 1, 1.41, 5.

Solution 7A.2

Pairwise distances are the same as before.
Step 1: Merge A and B (distance 1). Clusters: {AB}, {C}, {D}.
Complete-link distances:

$$d(AB, C) = \max(5, 5) = 5,$$

$$d(AB, D) = \max(7, 6) = 7,$$

$$d(C, D) = 1.41.$$

Step 2: Merge C and D (1.41). Clusters: {AB}, {CD}.
Distance:

$$d(AB, CD) = \max(5, 7, 5, 6) = 7.$$

Step 3: Merge {AB}, {CD}. Final distance = 7.

Solution 7B.1

Compute distances:

$$d(A,B) = 1,\ d(C,D) = 1.41,\ d(A,C) = \sqrt{41} = 6.4,\ d(A,D) = \sqrt{61} = 7.8,$$
$$d(B,C) = \sqrt{32} = 5.66,\ d(B,D) = \sqrt{52} = 7.21.$$

Largest distance:

$$d(A,D) = 7.8.$$

Split into:

$$\{A\} \quad \text{and} \quad \{B,C,D\}.$$

Next split cluster {B,C,D} using maximum distance inside it:

$$d(B,D) = 7.21 \quad \text{(largest)}$$

Split:

$$\{B\}, \qquad \{C,D\}.$$

Final clusters:

$$\{A\},\ \{B\},\ \{C,D\}.$$

Solution 7B.2

Cluster mean:

$$\mu = (5,5).$$

Distance to mean:

$$d(P_1,\mu) = \sqrt{(1-5)^2 + (2-5)^2} = \sqrt{25} = 5,$$
$$d(P_2,\mu) = \sqrt{(2-5)^2 + (1-5)^2} = \sqrt{25} = 5,$$
$$d(P_3,\mu) = \sqrt{(8-5)^2 + (9-5)^2} = \sqrt{25} = 5,$$
$$d(P_4,\mu) = \sqrt{(9-5)^2 + (8-5)^2} = \sqrt{25} = 5.$$

Try splitting into two obvious groups:

$$\{P_1,P_2\}, \qquad \{P_3,P_4\}.$$

Each subcluster has low SSE. Thus final clusters:

$$\{(1,2),(2,1)\}, \quad \{(8,9),(9,8)\}.$$

Solution 8.1

Distances:

$$d_1 = \sqrt{(2-1)^2 + (2-1)^2} = \sqrt{2},$$
$$d_2 = \sqrt{(2-5)^2 + (2-5)^2} = \sqrt{18}.$$

Formula:

$$u_i = \frac{1}{\sum_k (d_i/d_k)^{2/(m-1)}}.$$

Since $m = 2$, exponent $= 2$.

$$u_1 = \frac{1}{1 + (d_1/d_2)^2} = \frac{1}{1 + \frac{2}{18}} = \frac{1}{1.111} = 0.9.$$

$$u_2 = 1 - u_1 = 0.1.$$

Solution 8.2

Here,

$$u_{11}^2 = 0.64, \quad u_{12}^2 = 0.16.$$

Numerator:

$$0.64(0,0) + 0.16(2,2) = (0.32, 0.32).$$

Denominator:

$$0.64 + 0.16 = 0.80.$$

$$C_1 = (0.4, 0.4).$$

Solution 9.1

$$a(A) = d(A, B) = 1.$$

Distance to cluster 2:

$$b(A) = d(A, C) = 10.$$

Silhouette:

$$s(A) = \frac{b - a}{\max(a, b)} = \frac{10 - 1}{10} = 0.9.$$

Solution 9.2

Distance:

$$d(C_1, C_2) = \sqrt{32} = 5.657.$$

$$\frac{S_1 + S_2}{d} = \frac{3}{5.657} = 0.53.$$

With two clusters, DBI $= 0.53$.

Solution 10.1

Two visually distinct groups appear:

Red shades:

$$(255, 0, 0),\ (250, 10, 5)$$

Green shades:

$$(0, 255, 0),\ (5, 240, 10)$$

Using $k = 2$, the cluster centroids become:

$$C_1 = \left(\frac{255 + 250}{2}, \frac{0 + 10}{2}, \frac{0 + 5}{2}\right) = (252.5, 5, 2.5),$$

$$C_2 = \left(\frac{0 + 5}{2}, \frac{255 + 240}{2}, \frac{0 + 10}{2}\right) = (2.5, 247.5, 5).$$

All original pixels are replaced by their centroid color. Thus clustering reduces the color palette from 4 colors to 2 while preserving visual structure.

Solution 10.2

Four points form a tight cluster near:

$$\mu = \frac{1.2 + 1.0 + 1.3 + 0.9}{4} = 1.1.$$

The remaining point:

$$12.0$$

is far from this cluster (distance ≈ 10.9).

Thus clustering forms:

$$\{1.2, 1.0, 1.3, 0.9\} \quad \text{(normal behavior)},$$
$$\{12.0\} \quad \text{(anomaly)}.$$

Clustering separates regular customers from unusual high-value transactions, aiding fraud detection.

Solution 10.3

Compute natural groupings:

Action/comedy lovers (high ratings):

$$U_1 = (4,5), \quad U_2 = (5,4)$$

Low-rating users:

$$U_3 = (1,1), \quad U_4 = (2,1)$$

Thus clusters:

$$\{U_1, U_2\} \quad \text{(active, high-rating watchers)}, \qquad \{U_3, U_4\} \quad \text{(low-engagement users)}.$$

Recommendation benefit: Users in the same cluster receive similar movie suggestions, improving personalization.

Solution 10.4

Two groups naturally emerge:

Normal range:

$$(120, 180),\ (118, 175)$$

High risk (both high BP and high C):

$$(160, 240),\ (165, 250)$$

Clustering helps categorize patients into "normal" and "high-risk" groups, enabling doctors to prioritize screenings or interventions.

Solution 10.5

Two clusters are immediately visible:

Cluster 1 (downtown):

$$(2,3), (3,2)$$

Cluster 2 (highway region):

$$(20,25), (22,24), (19,26)$$

Authorities can use this clustering to allocate police patrols or redesign traffic flow.
